

# A COMBINED SEA TEMPERATURE NETWORK FOR THE COASTAL SOUTHEASTERN UNITED STATES

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## Abstract

Across a region spanning from the West Florida Shelf, around the Keys, and through the Atlantic Ocean to the North Carolina border with Virginia (the SECOORA region), nearly 1,000 locations have measured sea water temperatures (SWTs) over the past 50 years. We investigated the SWTs for the occurrence of marine heatwaves (MHWs) and for climate trends. Across the region, there has been an increase in both the frequency and intensity of MHWs along with a rise in SWTs that is greater than the global average, with the median trend across the region being 0.47°C/decade. During the past 10 years, many regions have experienced the most intense MHWs in their period of record. The FACT Network, a system of subsurface SWT probes, was shown to be the start of an important addition to the SWT record. Subregions lacking routine *in situ* SWT measurements were identified as were locations where adding new subsurface measurements or maintaining the preexisting measurements would aid physical and biological marine research projects.

## 1. Introduction

### 1.1 Water Temperature Monitoring

Water temperature monitoring was a basic component of ships logs starting in the late 1600s (e.g. Slutz et al., 1985; Woodruff et al., 1998). After World War 2, the monitoring of temperatures in the global oceans began to be expanded with buoys used as continuous monitoring systems with the resultant data arrays providing a method for climate analysis (e.g. Vose et al., 2012) and for bias-correcting satellite measurements (Lazarus et al., 2007). Since then, *in situ* measurements from buoys have provided important baselines for monitoring climate change (e.g. Nickerson et al., 2023) and for evaluating model performance during tropical cyclones (e.g. Liu et al., 2025; Nickerson et al., 2025). The difficulty in using *in situ* data is that measurement errors vary from instrument to instrument across both time and space (Smith and Reynolds, 2002, 2003, 2005; Reynolds et al., 2002, 2007; Kent and Challenor, 2006; Kennedy et al., 2011; Vose et al., 2012; Huang et al., 2015, 2017).

#### *1.1.1 Integrated Ocean Observing System and Regional Associations*

The Integrated Ocean Observing System (IOOS) was formally established in 2011 to put oceanic data encapsulating various federal partners, observation networks, and ocean variables into one, large catalog where the data can be readily accessed by those who need it. Specifically, there are seven goals that the IOOS program intends to meet related to the health and safety of the public and the conservation of natural resources and environments. The IOOS system consists of 11 regional associations that each contribute data to a single web portal while also maintaining their own individual web portals. This can data can be found at <https://data.ioos.us/>

From these regional associations, numerous variables are routinely collected and uploaded to the system, including sea water temperature (SWT). Among the regional associations, the Southeast Coastal Ocean Observing Regional Association (SECOORA) stretches from Florida (starting at the Alabama border) to the border between North Carolina and Virginia; the Gulf of America Coastal Observing System (GCOOS) proceeds west from Alabama to the Mexican border; and the Caribbean Coastal Observing System (CariCOOS) encompasses Puerto Rico and the U.S. Virgin Islands. Across these three regions are over 900 *in situ* water quality monitoring stations which include SWT among the variables collected. Many of these stations transmit in real time as well, with temperatures available through either the IOOS site or the appropriate regional site.

### *1.1.2 FACT*

Among the *in situ* water quality sites numbered above are over 600 stations that measure SWT in the SECOORA and CariCOOS regions as part of the FACT Network, which extends from Puerto Rico to Virginia. The FACT Network (Young et al., 2020) began in 2007 and now includes over 900 bottom-mounted acoustic receivers (across nearly 100 partner groups) for monitoring the movement of tagged species, with many of these including a temperature sensor. Such stations do not provide real-time data; the consequences of this are explained in Section 2.2.2.

### *1.1.3 National Data Buoy Center*

In addition to the data downloaded from the IOOS site, there are numerous SWT measurements from the National Data Buoy Center (NDBC) and their many partner organizations/institutions. These stations are scattered across the globe, but this analysis is focused on 127 stations located in the Western Atlantic Ocean, the Caribbean Sea, and the Gulf of America. All of these stations provide near real-time surface temperature measurements.

## 1.2 Marine Heat Waves (MHWs)

### *1.2.1 General overview*

Marine heat waves (MHWs) are analogous to atmospheric heat waves (e.g. Hobday et al., 2016). Meteorologically, MHWs have been shown to influence the monsoon patterns over the Indian subcontinent (Saranya et al., 2022) and intensify tropical cyclones (Singh et al., 2021; Mawren et al., 2022; Rathore et al., 2022; Roxy et al., 2024). Biologically, MHWs cause mass mortality events in marine species (Smith et al., 2021; Jones et al., 2018; Garrabou et al., 2022; Smith et al., 2023), initiate coral bleaching (Eakin et al., 2019; Moriarty et al., 2023), and can alter local environments (Smith et al., 2024; Filbee-Dexter, et al., 2020; Smale et al., 2019; Wernberg et al., 2013; Stuart-Smith et al., 2018).

The definition of a MHW (Hobday et al., 2016) requires that they be (1) anomalously warm, (2) prolonged, and (3) discrete. We summarize these three definitions.

- 1) Anomalously warm: The temperature is higher than a chosen percent threshold based on an 11-day window centered on the date from which the mean and threshold are computed and using all data over a chosen timeframe. A typical threshold is the 90th percentile.
- 2) Prolonged: The events must last at least five days
- 3) Discrete: The events must be separated by at least three days.

### 1.2.2 More specific regional impacts

The definition of an MHW given above constrains the datasets which can accurately estimate the occurrence of such an event. For climatology, a minimum of 30 years of data is needed in order to resolve the major global processes without biasing the climatology. Hence, full understanding of the spatial extent of MHWs necessitates the consideration of both short and long time series.

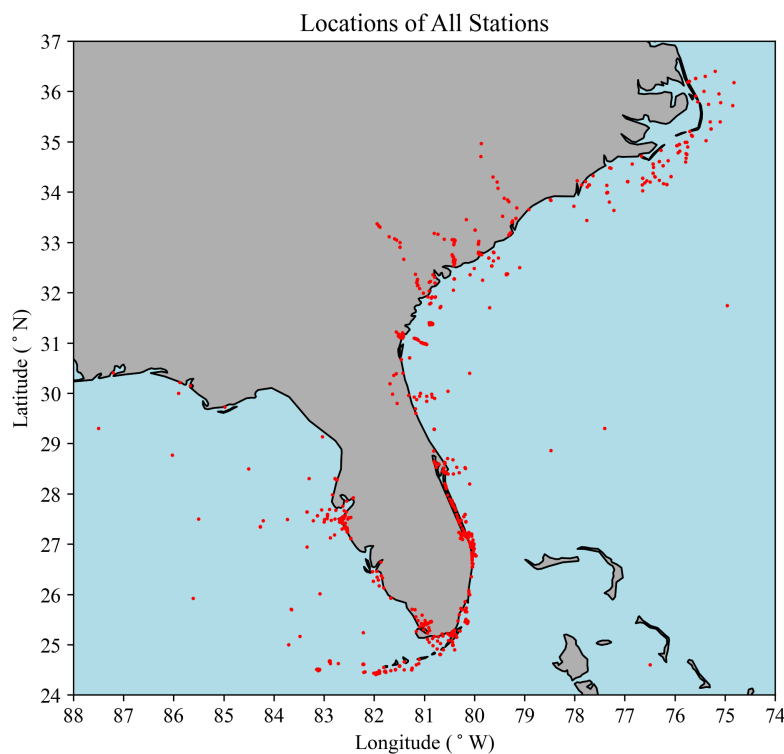


Figure 2.1: Map showing the locations of all stations used in this study. Stations shown are from the IOOS & SECOORA ERDDAP servers and from the NDBC website.

### 1.3 Purpose of This Report

This report will analyze the SWT trends from nearly 1000 stations throughout the coastal waters in the SECOORA region.. Quality control (QC) methods will be established in keeping with the QARTOD concepts established by IOOS (Bushnell et al., 2018). Sparse data necessitate further analyses in order to make meaningful inferences of climate patterns (e.g. Nickerson et al., 2023).

Section 2 explains the classification of the SWT measurement stations and the QC methods that were applied to the data. The methodology details the entire process from downloading the data to performing the MHW analyses.

Section 3 will provide classic statistics (trends, trend errors where applicable) for individual time series as well as comparisons between adjacent stations (T-tests, F-tests, etc.) while contextualizing these in an MHW-focused framework. Section 4 will finally provide an analysis of the results focused on the FACT network and SECOORA.

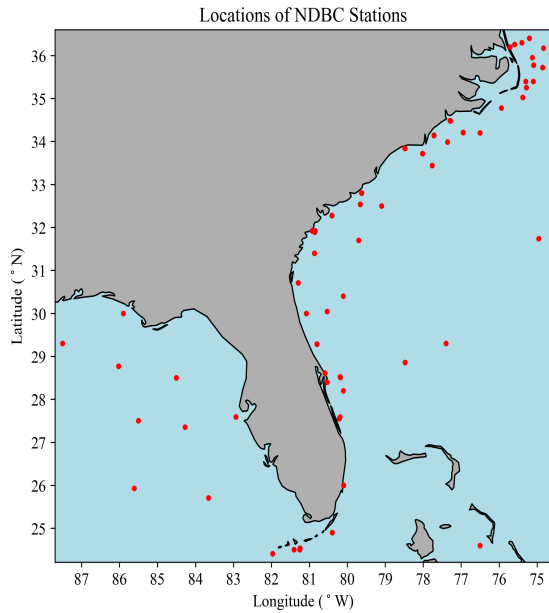


Figure 2.2: Map showing the locations of all data acquired from the NDBC website.

## 2. Data

### 2.1 Breakdown of stations

For purposes of this analysis, we separated the stations into four major classifications: FACT SWT probes, moored buoys whose data archives are hosted by the NDBC, moored buoys from the University of South Florida Coastal Ocean Monitoring and Prediction Systems (USF COMPS), and other SWT stations that did not fit any of the previous groups. For some analyses, these stations (Figure 2.1), were grouped by location, with the location-based groups being stations (1) on the West Florida Shelf, (2) in the Everglades or the Florida Keys, (3) eastern coastal waters of Florida, (4) coastal waters of Georgia and South Carolina, and (5) coastal waters of North Carolina.

#### 2.1.1 NDBC & Partners

The NDBC provides real-time data at locations scattered across the Pacific and Atlantic Ocean and the Great Lakes, with approximately 200 buoys maintained by the NDBC and various partner organizations across these bodies of water. In the SECOORA region, there are 65 buoys whose data are stored on the NDBC web server (Figure 2.2). For the actual NDBC buoys, the temperature measured by the thermistor is from the hull temperature as measured 0.6-1.1 m below the sea surface with an accuracy of  $\pm 1.0$  °C. No QC is performed on these data before they are uploaded to the NDBC servers.

#### 2.1.2 FACT

FACT probes are non-real-time typically bottom-mount sensors associated with acoustic telemetry studies used to monitor the movement of tagged fauna (Young et al., 2020). The data are continuously logged at regular intervals until the data are downloaded from the probe following instrument retrieval. Of nearly 1000 FACT acoustic telemetry probes located in the Atlantic, Gulf, and Caribbean, over 600 of these include temperature measurements (Figure 2.3) from 20 institutions. This report provides the first look into the data collected from these probes and performs the first QC on them, with focus on those located within the SECOORA region.

#### 2.1.3 USF-COMPS

COMPS data were uploaded to ERDDAP with QC measures having already been performed on all of the data. Instrument design ranges, local climatology, and data filters are used to flag values. Temperature measurements at the surface and at intermediate depths come from sensors

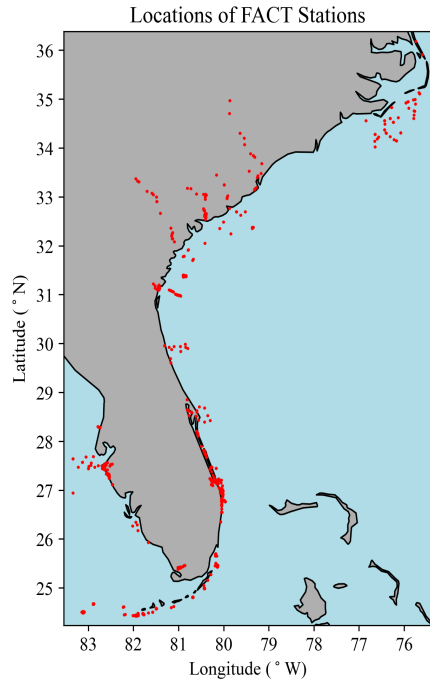


Figure 2.3: Map showing the locations of the FACT probes

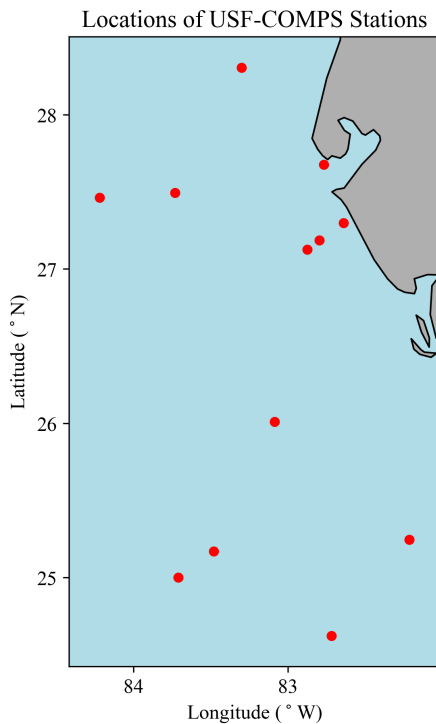


Figure 2.4: Map showing the locations of all USF-COMPS stations

attached to the buoy bridle (stations C10, C12, C13, C14, C16, C17, C21, and C22) while bottom temperatures come from bottom-mounted Acoustic Doppler Current Profiler (ADCP) (Nickerson et al., 2023; Sorinas et al., 2023). Stations C11, C15, C18, and C19 only have bottom-mounted ADCPs without any measurements from the surface or the intermediate water column.

#### 2.1.4 Other stations

The remaining stations are scattered throughout all three main bodies of water that are the focus of this analysis. These data come from a variety of instruments, but all of them are surface data. Stations and data arrays at 284 locations from 18 institutions contributed to this subset of data.

## 2.2 Methods

### 2.2.1 Data collection

Data collection began with splitting the process into three streams: one for the IOOS ERDDAP, one for the SECOORA ERDDAP, and one for the NDBC website. While some NDBC partner buoys do have listings on the IOOS ERDDAP, these time series do not cover the entire time series. This is also the case with the splitting of the IOOS and SECOORA ERDDAPs. Many stations are listed twice on the IOOS server, with one of those being the full dataset and the other being a shortened version of that data set (much like the case with the NDBC partner buoys).

Stations were limited to a box encompassing the Gulf, the Caribbean and the SECOORA region (87.5°W to 74°W and 22°N to 36.5°N). As part of each ERDDAP server, there is a master list that contains the most important station information: latitude and longitude ranges, station name and owner, etc. Riverine stations, drifters, and gliders were excluded from analysis, except for FACT

probes that were located in rivers. Most stations measured in hourly intervals, but many measured SWTs at higher frequencies.

NDBC partner buoy locations were identified by a number, with 5-digit integers in the numerical format 41xxx representing Atlantic buoys, 42xxx representing Gulf buoys, 44xxx representing the Atlantic spanning from North Carolina to the northeastern United States. The data are stored in a series of annual text files, where each file contains one year of data. Most buoys measured SWTs in hourly intervals until 2005, when some stations began to measure more frequently.

Due to the volume of data that is downloaded from each server, all of the codes were written in a manner that makes it easy to add clauses to resume the data download should the code somehow be interrupted. Once downloaded, NDBC annual text files were combined into single netCDF files that contained all data for that station. All data downloaded from ERDDAP servers were also in netCDF format. The ReadMe file provides extra details about downloading the data.

| Table 2.1: QARTOD flag values and their meanings |                                       |
|--------------------------------------------------|---------------------------------------|
| Flag                                             | Meaning                               |
| 1                                                | Passed quality control: ready for use |
| 2                                                | No quality control performed          |
| 3                                                | Suspect data: only use with caution   |
| 4                                                | Failed quality control: do not use    |
| 9                                                | No data                               |

### 2.2.2 *Quality control*

QC methods were developed according to the QARTOD manual (Bushnell et al., 2018) and the methods outlined by Nickerson et al. (2017, 2023) for processing data from moored stations. All QC methods are chosen subjectively, and they can be classified based on whether the method is objective or subjective. Subjective methods should only be used when objective methods fail. Table 2.1 outlines the QARTOD

flags and a brief explanation of what they mean. The methods described in this section were applied to all time series except for the UFS COMPS data, which already were QC'd before the data was uploaded to the ERDDAP server.

The first test applied is the spike test. This test checks for sudden changes in the SWT that are outside physically plausible values. Due to most instruments having a standard error of  $\pm 1^\circ\text{C}$  and the realistic possibility of oceanic fronts, upwelling/downwelling, or storms leading to large and sudden changes in SWT, large changes from one measurement to the next (especially for hourly measurements) can plausibly occur. Hence, we applied two spike tests,

$$|0.5*(T_{j+1} + T_{j-1}) - T_j| > 3 \quad (1),$$

$$|0.5*(T_{j+1} + T_{j-1}) - T_j| > 8 \quad (2),$$

and assigned a QARTOD flag 3 if the value failed the test in equation (1) and flag 4 if the value failed the test in equation (2). The threshold values 3 and 8 were arbitrarily chosen to maintain consistency with the QARTOD manual.

The second test is the standard deviation test. There are multiple layers to this test to ensure that only truly suspect data is flagged. First, outliers are flagged relative to the mean SWT over a full 25-hour running mean (one full lunar day) across the entire time series, for which also computed the standard deviation,  $\sigma_{25}$ . SWTs are flagged only if they fail two different tests,

$$|T_j - T_{25}| > 2 + 3*\sigma_{25} \quad (3),$$

which is the standard QARTOD standard deviation test, and if they're outside the 99.7 quantile,  $Q_{25}(99.7)$  and  $Q_{25}(0.3)$ . If the value fails *both* tests, then it's assigned flag 3. We repeated the test for the 30-day (720-hour) running means,  $T_{720}$ , standard deviation,  $\sigma_{720}$ , and quantiles,  $Q_{720}$ .

The methods applied in the standard deviation test were repeated to determine if the SWT at any time  $T_j$  falls within the standard threshold for *monthly* data over the time series. On a month by month basis, all measurements during a given month had their mean,  $T_M$ , and standard deviation,  $\sigma_M$ , computed and compared using the same test,

$$|T_j - T_M| > 2 + 3*\sigma_M, M = 1, 2, \dots, 12 \quad (4).$$

Points that failed the test outlined in equation (4) and quantiles  $Q_M$  were assigned flag 3 for being outside the local monthly climatology. We also passed through each individual month within an individual year and applied the same method, checking for the mean,  $T_D$ , standard deviation,  $\sigma_D$ , and quantiles,  $Q_D$ .

The next objective test checked for intermittency. Over the course of a 720-hour rolling window, values were flagged if there were 5 or fewer measurements during that rolling window. Points were then assigned flag 4 due to the likelihood of instrument failure.

Following this was the objective test for local climatology. In some regions, temperatures never exceed certain thresholds or go below certain thresholds. These thresholds must be set carefully, especially in a changing climate, to ensure that a future application of such methods does not flag good data. An example is the Gulf Star Marina near Fort Myers, Florida, where temperatures are never less than 15°C. Hence, these values are flagged for falling outside the local climatology with flag 3.

Lastly, there were subjective methods of data curation which necessitated specific attention to the data on a station-by-station basis. An example of such extreme scenarios is shown in Figure 2.5, where the probe appears to be recording common indoor air temperatures, due to FACT probes continuing to automatically log data from initiation until either (1) the data is downloaded or (2)

the battery fails. Other subjective methods include using immediately adjacent stations to identify local outliers not detected by the previous methods.

### 2.2.3 Climatology

Climatologies were computed following the methods described in Nickerson et al. (2023), with the monthly mean SWTs only being computed if there were no more than 7 days (168 hours) of empty or flagged data points during an entire month. Likewise, annual means were computed if there were at least 10 months for which monthly means were computed during that year.

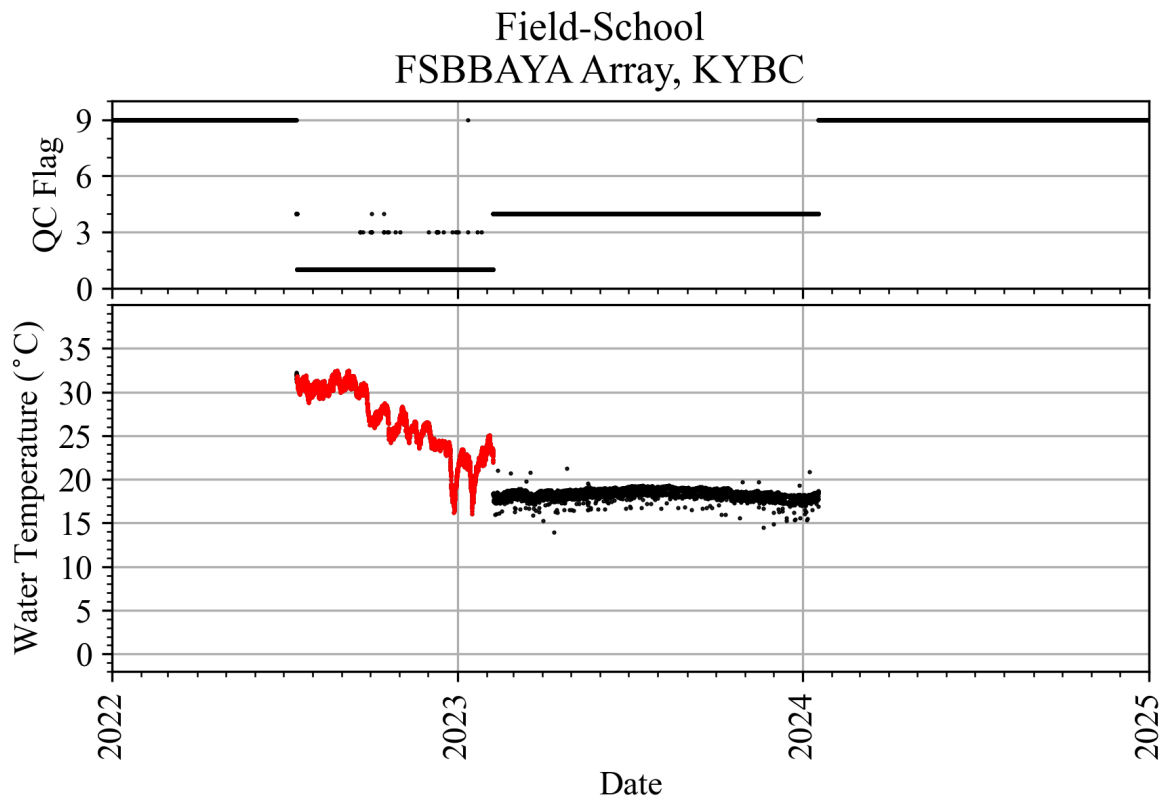


Figure 2.5: (Top) QARTOD flags assigned to each time step. Note that the time series was expanded to complete the calendar years in order for simple application of the QC methods that were described in section of 2.2.2. (Bottom) QC'd sea water temperatures (red) and flagged temperatures (black) at a FACT station with an extended period of questionable data.

Linear trends were computed for all stations having at least 3 years with annual means, but trend statistics (degrees of freedom, 95% confidence intervals) were only computed for stations with at least 10 years with annual means. While consideration of the cyclic nature of systems provides a more accurate estimate of the degrees of freedom in a climate system (Nickerson et al., 2023), for simplicity this analysis uses the classic method of 2 less than the number of points.



| Table 2.2: Marine heat wave categories and their categorically assigned values |            |        |
|--------------------------------------------------------------------------------|------------|--------|
| Category                                                                       | Definition | Weight |
| Moderate                                                                       | $1-2*T_h$  | 1      |
| Strong                                                                         | $2-3*T_h$  | 2      |
| Severe                                                                         | $3-4*T_h$  | 3      |
| Extreme                                                                        | $> 4*T_h$  | 4      |

#### 2.2.4 Marine Heat Waves (MHWs)

In this analysis, MHWs were computed according to the methodology described in Hobday et al. (2016) and as implemented into Python. The code can be downloaded from GitHub at the following:

<https://github.com/ecjoliver/marineHeatWaves> For this analysis, a few minor updates were made to the Python code in order to bring it up to more modern standards. The threshold for a MHW is defined as the difference between the local climatological mean and the 90th-percentile, which we define as  $T_h$ . To each MHW, we assign a weight to compare the MHW intensities. The definitions of the

categories and their weights are shown in Table 2.2.

### 2.3 GitHub

All codes and functions were written in Python 3 and are available in the SECOORA GitHub repository: <https://github.com/SECOORA/marine-heat>. Included in the repository is a ReadMe file that details the purpose of each of the codes and the proper order in which to run the codes. By running the code as provided, the figures in this analysis can be [approximately] duplicated. Differences will be seen as a result of (1) new data being added to the ERDDAP servers and (2) new stations being added to the same servers.

## 3. Results

### 3.1 Basic Statistics

Despite there being nearly 1000 stations with data included in this analysis, this basic approach to SWT analysis results in only one station having at least 30 years in which the annual mean was computed despite 8 stations having more than 30 years of data (Figure 3.1, Table 3.1). Of the 8 stations with more than 30 years of data, all but one are NDBC partner buoys, and the other is part of the NOAA Center for Operational Oceanographic Products and Services (CO-OPS). These are all located in the open waters of the Atlantic and the Gulf.

The majority of stations (over 600 out of nearly 1,000 stations) have 2 or fewer years with annual mean water temperature measurements. There were 35 stations in the 20-30 year range, which is on the cusp of the 30-year threshold for computing climatologies that resolve the major global cycles (e.g. Nickerson & Maul, 2017). Nevertheless, nearly 150 stations have at least 10 years of data, which is enough to at least see major patterns emerging (e.g. Nickerson et al., 2023). Lack of funding, weather events, collisions with vessels, and instrument failure can lead to the removal of moored instruments (e.g. Nickerson & Maul, 2017).

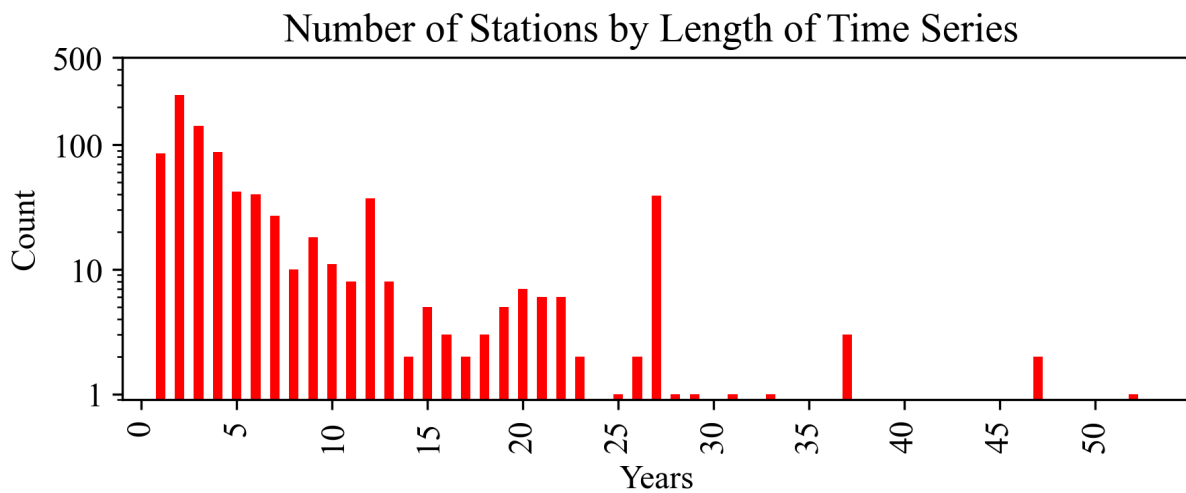


Figure 3.1: Stations by length of time series (in years). Note the logarithmic y-axis scale.

Table 3.1: Counts of stations by the length of the time series (in years) and the number of annual means computed at each station.

| Range       | Time Series Length | Annual Means |
|-------------|--------------------|--------------|
| <1 year     | 0                  | 339          |
| 1-2 years   | 334                | 290          |
| 3-5 years   | 228                | 57           |
| 6-10 years  | 106                | 74           |
| 11-15 years | 60                 | 14           |
| 16-20 years | 20                 | 19           |
| 21-30 years | 58                 | 35           |
| 31-40 years | 5                  | 0            |
| 41-50 years | 2                  | 0            |
| >50 years   | 1                  | 0            |

### 3.1.1 Annual Summaries

With the passing of time, more and more stations have been established, such that the number of fixed SWT measurements in the SECOORA region has increased from just 2 stations collecting

enough data to compute an annual mean in 1980 to over 100 stations with such data from 2020 to 2024 (Figure 3.2). The decrease in 2025 can be at least partially attributed to fixed stations that do not automatically transmit their data to ERDDAP. Most stations with subsurface sensors do not have automatic transmitters, and the data must be downloaded directly from the sensor.

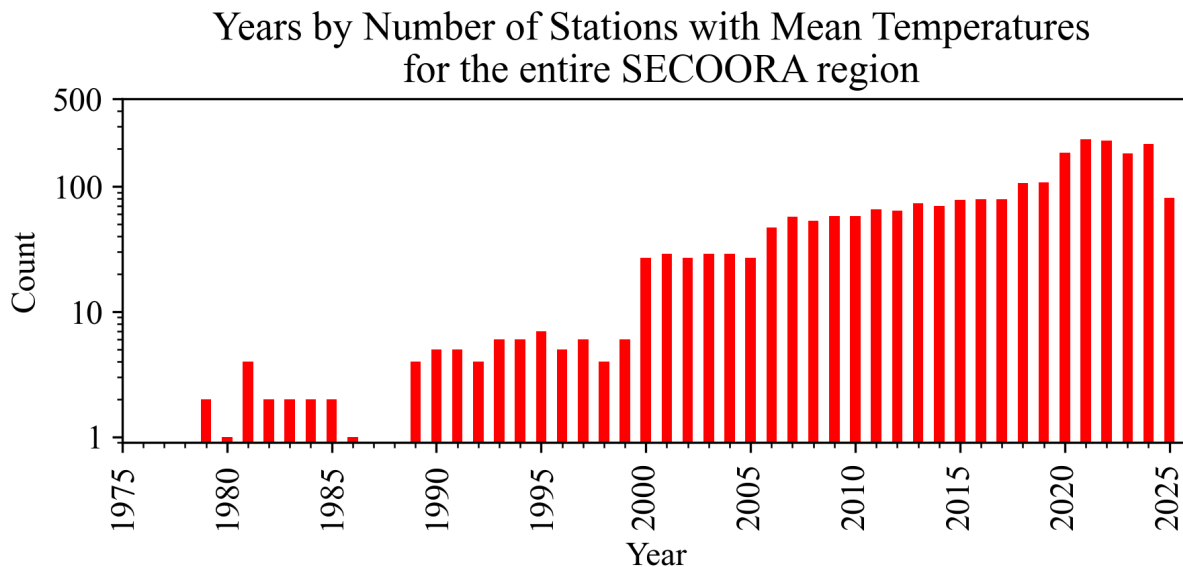


Figure 3.2: Bar chart showing the number of stations measuring SWTs in each year.

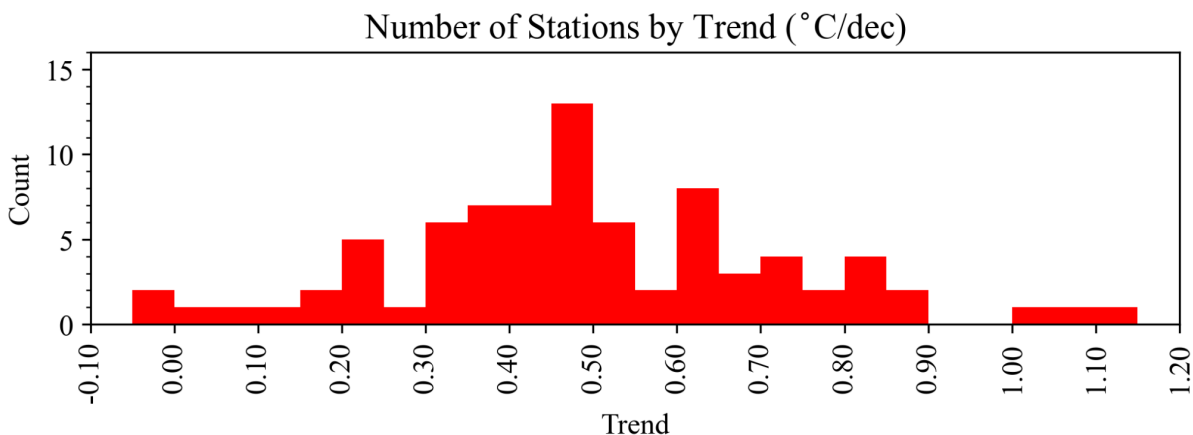


Figure 3.3: Bar chart showing the number of stations with trends within bins of  $0.05^{\circ}\text{C}/\text{decade}$ . No single year had at least 10 stations active in the entire SECOORA region until the 2000, with this including several NDBC partner stations, some of the USF-COMPS buoys, and one NOAA CO-OPS buoy. SWTs have, on average, steadily increased over the period of record, with the stations showing trends ranging from only slightly negative to over  $1^{\circ}\text{C}/\text{decade}$  (Figure 3.3). While extremes can be a result of biased trends (Nickerson et al., 2017), the majority of stations

with the prerequisite 10 annual means have trends in the range 0.3-0.9°C/decade, with the region warming at a median rate of 0.47°C/decade for all stations.

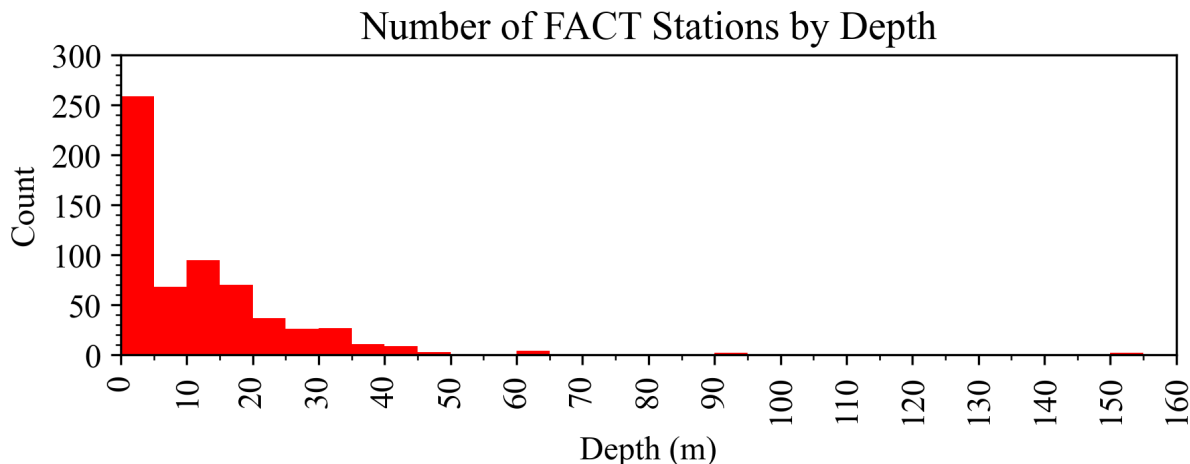


Figure 3.4: Bar chart showing the number of FACT stations with depths in bins of 5 meters.

Subsurface fixed-instrument SWT measurements are limited to those from the USF COMPS program and from the FACT network. These measurements span from the middle of the water column to bottom measurements at the COMPS stations with mostly near-bottom measurements for the FACT sensors. While COMPS is limited to the West Florida Shelf, the FACT network has sensors across the entire SECOORA region with measurements ranging from very shallow water (less than 5 m depth) to over 150 m beneath the surface (Fig. 3.4)4. Presence of MHWs in SE US Coastal Waters

The most difficult part of a MHW analysis can be finding a climatology to use with stations that have varying record lengths (Fig. 3.1; also see Hobday et al., 2018). All results must be weighed in light of these differences, but we note that normalization and standardization do not change the results described herein. Hence, in a region warming as rapidly as the SECOORA region, the detection of a heatwave at a location with a shorter time series represents only a minimum due to the lack of data in earlier, cooler years.

We start by examining time series from the open waters of the Atlantic Ocean, the Atlantic shelf region, the Florida Everglades, and the West Florida Shelf. For simplified comparison, we used a common time range of 1995 to 2025 for plotting the data in these time series. Each of the next four figures (Figs. 4.1-4.4) shows a time series of SWTs at a station with the MHWs highlighted in blue (moderate), green (strong), red (severe), or magenta (extreme). These stations were chosen due to having a robust, long time series.

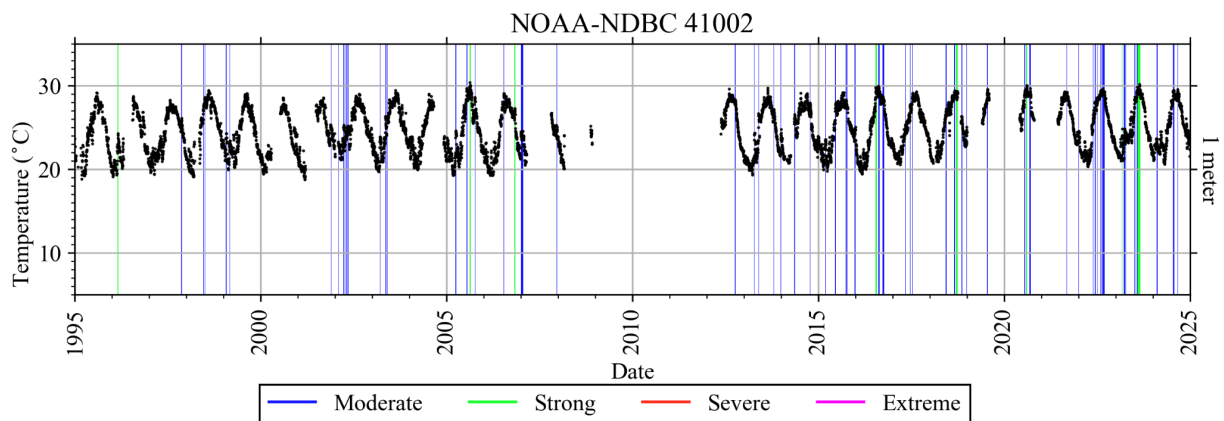


Figure 4.1: Sea water temperatures (black) at NDBC station 41002 with the heatwaves highlighted as moderate (blue), strong (green), severe (red), and extreme (pink)

At each of the stations, there is the recurring pattern of having more frequent and longer MHWs in more recent years. NDBC buoy 41002 (Fig. 4.1) represents one of the longest time series in the full dataset. Across this station and all other stations of similar length (e.g. NDBC 41008), we observe this same pattern of increasingly frequent, and stronger, MHWs. Most of the NDBC buoys (and many of their partner buoys) are located in the open ocean.

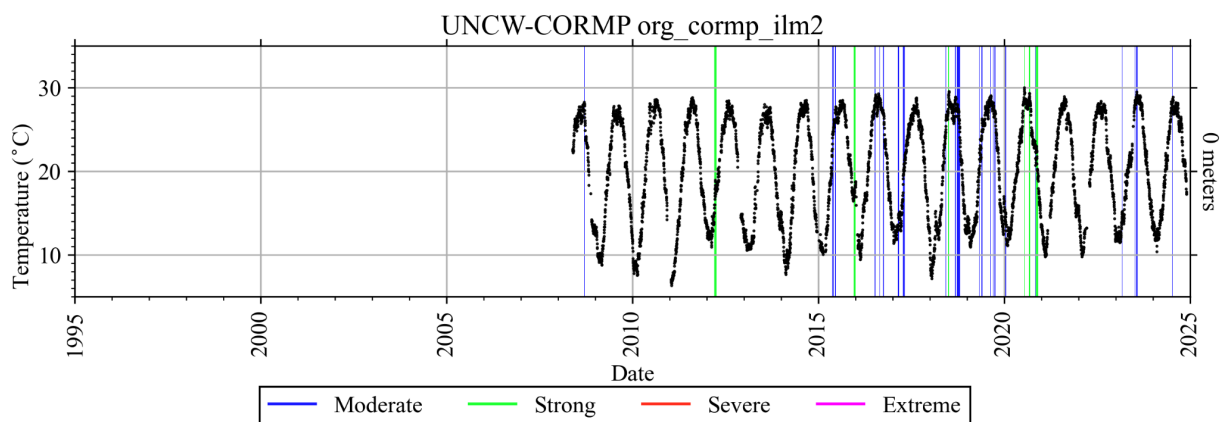


Figure 4.2: Sea water temperatures (black) at CORMP station ILM2 with the heatwaves highlighted as moderate (blue), strong (green), severe (red), and extreme (pink)

Inshore from NDBC 41002, at station ILM2 (Fig. 4.2), increased seasonal variability and cooler SWTs are observed, but there have been many MHWs in recent years. Across the ensemble of buoys maintained by the University of North Carolina Wilmington Coastal Ocean Research and Monitoring Program (CORMP), warming SWTs and increasingly frequent MHWs are seen. Of these stations, only SUN2, located near the 10 m isobath in the Carolina's Long Bay region, had a severe MHW in its period of record.

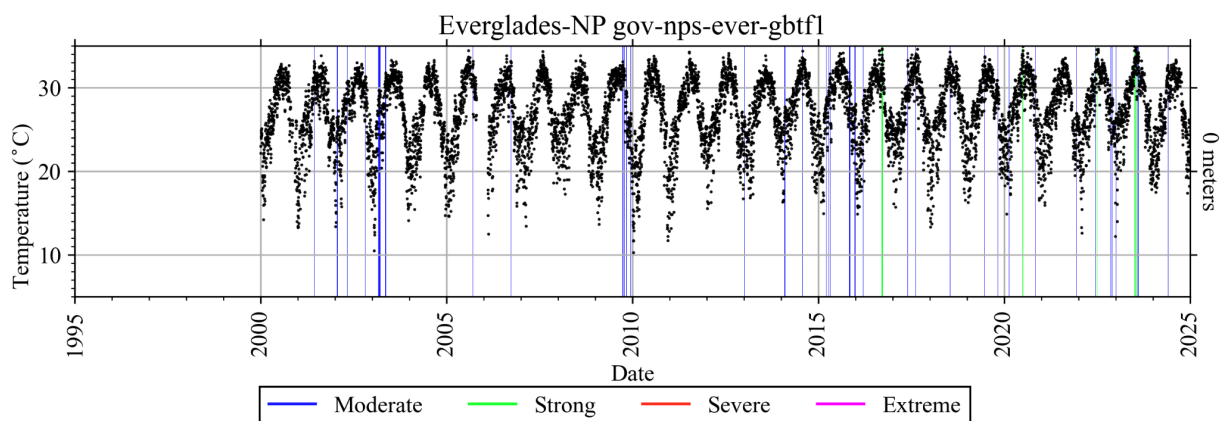


Figure 4.3: Sea water temperatures (black) at Everglades National Park station GBTF1 with the heatwaves highlighted as moderate (blue), strong (green), severe (red), and extreme (pink)

In this region is also one of the most glaring gaps in the data record, with Pamlico Sound having no continuous *in situ* observation systems being located within this region. The Pamlico Sound is the second largest lagoonal estuary in the United States (Paerl et al., 2007) and is important as a nursery habitat for many fish species (NCDEQ, 2016; Ross & Epperly, 1985) and is important for commercial and recreational purposes (NCDMF, 2017).

Even in the swamp environment of the Florida Everglades (Fig. 4.3), MHWs have occurred with increasing frequency in more recent years. The first severe MHW recorded at any stations in the network occurred in 2019. With many stations in this region measuring SWT trends greater than  $0.5^{\circ}\text{C}/\text{decade}$ , the impacts of MHWs might rapidly increase if the warming trends continue.

On the West Florida Shelf, station C10 (Fig. 4.4) has data for the entire water column, with SWT measurements at the surface, 4 meters depth, 10 meter depth, and at 19 meter depth (the bottom). At all depths, the frequency and intensity of MHWs has increased over the 26 year period of data collection, including, in 2023, a severe heatwave that lasted for over 6 weeks at both the surface and at 10 meters depth. At the bottom, the event signal is obscured by missing data.

The range in SWT trends indicates that there is a local variation component to MHW frequency, but the overall trend towards warmer temperatures across the SECOORA region is the primary driver in the increase in MHWs. With regards to MHWs, at stations where sufficient data exists, choosing a representative baseline period is essential for proper analysis of trends across spatial and temporal scales (Hobday et al., 2018).

At each station, we computed the number of days each year which were spent in MHWs of each category and then computed averages of these for all stations in the SECOORA region, on the West Florida Shelf, in the Florida Everglades and Keys, and in the open waters of the Atlantic to

the east of Florida, Georgia and South Carolina, and North Carolina. Using averages reduces any bias that may be introduced as a result of the increased number of stations in more modern years, as shown in Fig. 3.2 while still giving weight to MHWs such as the ones in Fig. 4.4. In addition to the categorical averages, we show the total number of MHW days for the given year. In the supplementary information are figures with the station counts by year for each region.

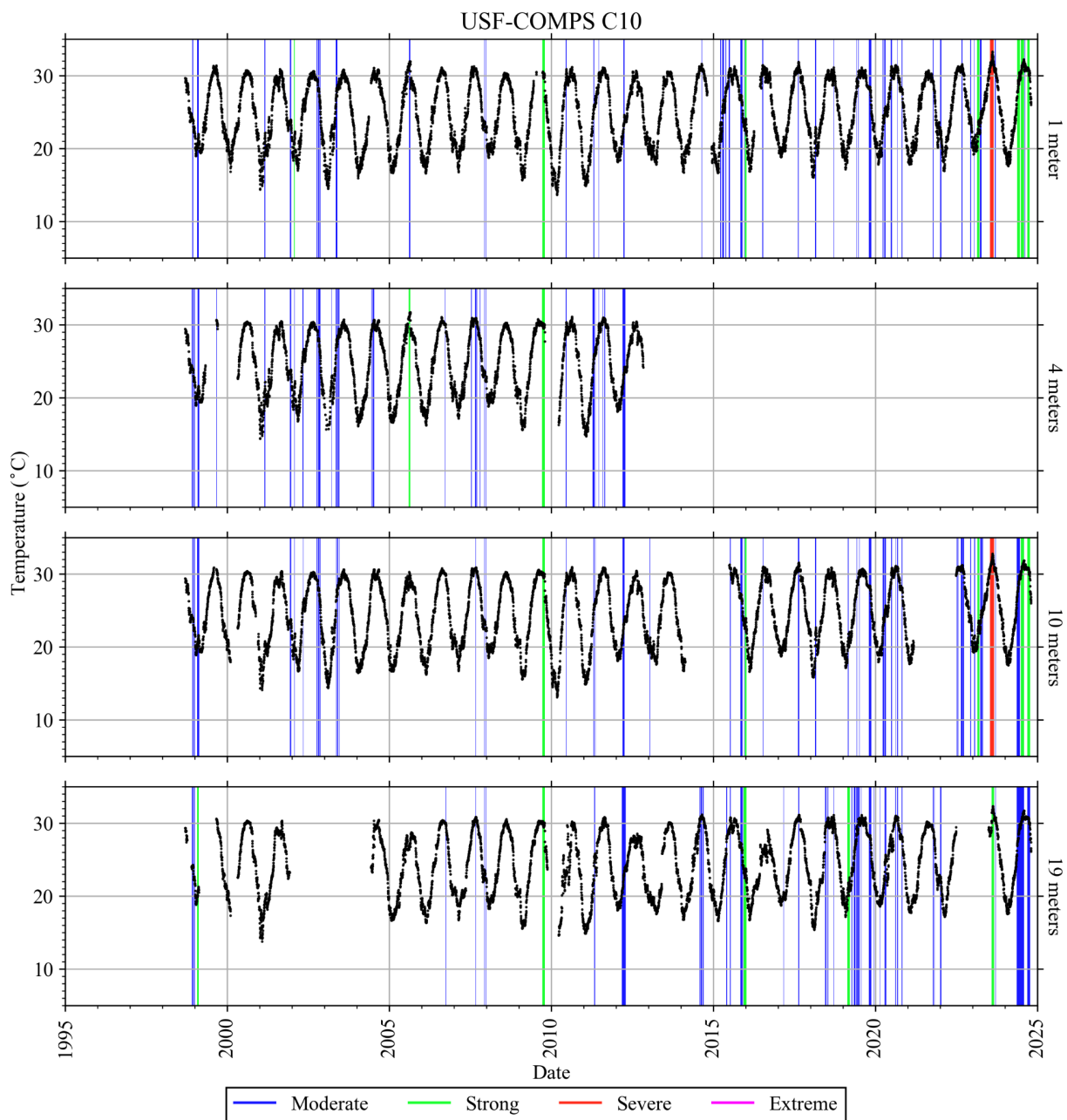


Figure 4.4: Sea water temperatures (black) at COMPS station C10 with the heatwaves highlighted as moderate (blue), strong (green), severe (red), and extreme (pink)

Throughout the SECOORA region, MHWs increased in frequency and intensity with time (Fig. 4.5). The first severe MHW in the SECOORA region did not occur until 2012, and the first

extreme MHW was in 2019. Over time, MHWs have become more frequent, with 1982, a year in which there was a strong El Niño, being the lone year before 2015 in which there were more than 20 MHW days, on average, at locations across the SECOORA region.

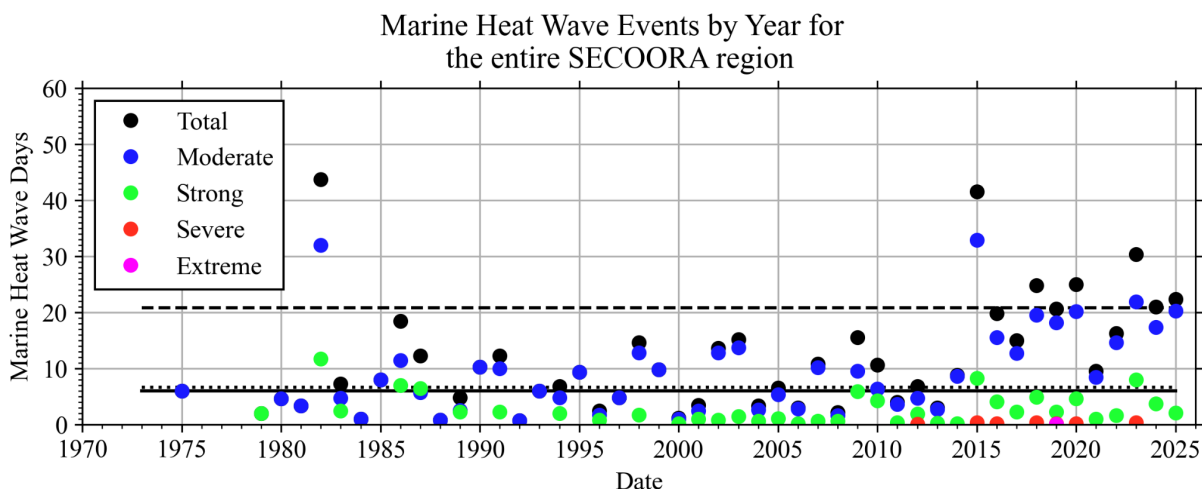


Figure 4.5: Mean Marine Heat Wave Days (black dots) per station by year in the SECOORA region with the categories highlighted as moderate (blue), strong (green), severe (red), and extreme (pink). The solid black line represents the median marine heat wave days for 1975-1995; the dotted black line is for 1995-2015; and the dashed black line is for 2015-2025.

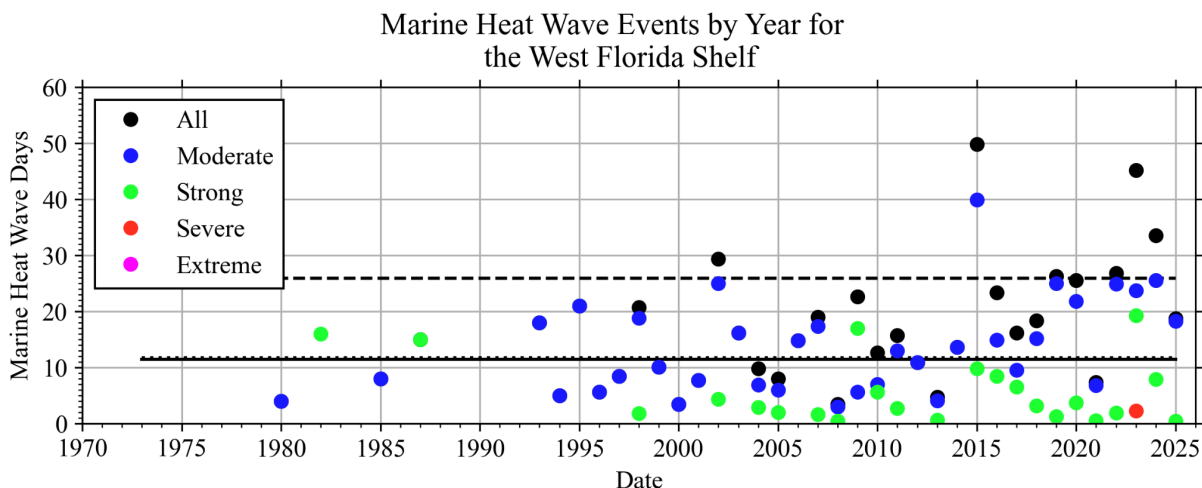


Figure 4.6: Mean Marine Heat Wave Days (black dots) per station by year in the West Florida Shelf region with the categories highlighted as moderate (blue), strong (green), severe (red), and extreme (pink). The solid black line represents the median marine heat wave days for 1975-1995; the dotted black line is for 1995-2015; and the dashed black line is for 2015-2025.

On the West Florida Shelf (Fig. 4.6), the pattern has been repeated, with MHWs becoming even more frequent, locally, throughout the period of record, with MHWs having top-to-bottom effects



in the water column (Fig. 4.3). While measurements began in 1980, a consistent, reliable record of SWTs only began in 1993. The region had its first severe MHW in 2023.

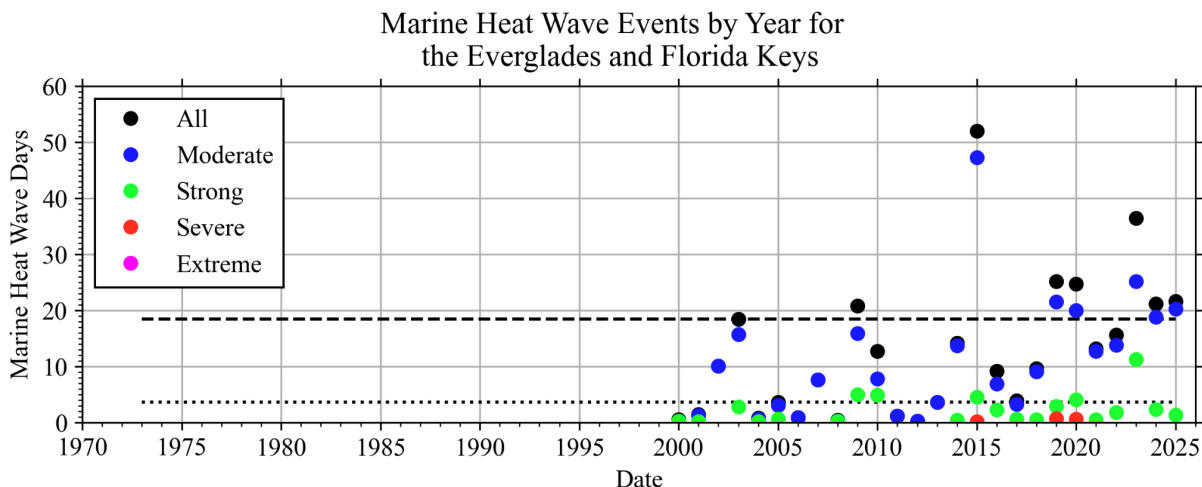


Figure 4.7: Mean Marine Heat Wave Days (black dots) per station by year in the Florida Keys and Everglades with the categories highlighted as moderate (blue), strong (green), severe (red), and extreme (pink). The solid black line represents the median marine heat wave days for 1975-1995; the dotted black line is for 1995-2015; and the dashed black line is for 2015-2025.

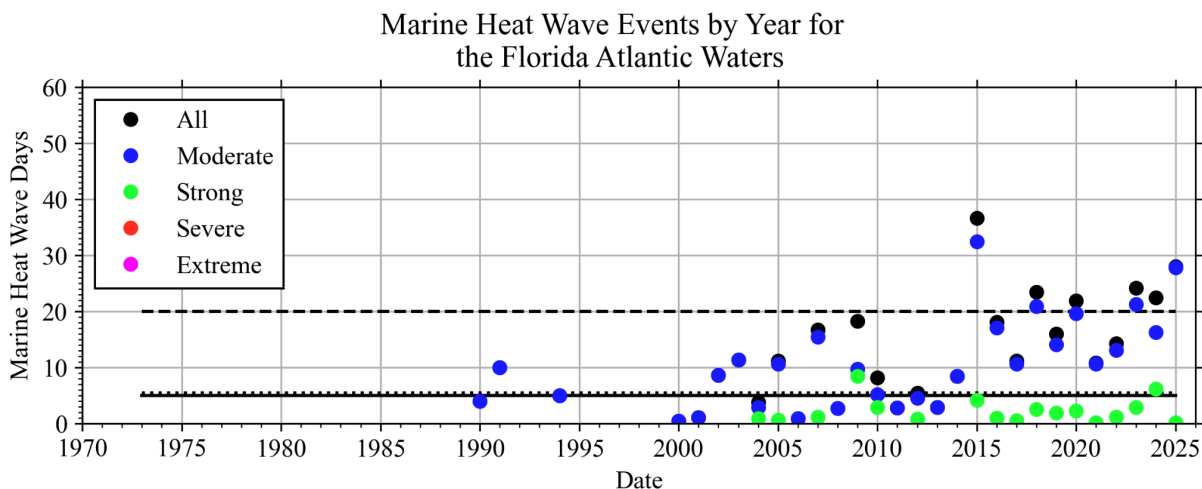


Figure 4.8: Mean Marine Heat Wave Days (black dots) per station by year in the Florida Atlantic waters with the categories highlighted as moderate (blue), strong (green), severe (red), and extreme (pink). The solid black line represents the median marine heat wave days for 1975-1995; the dotted black line is for 1995-2015; and the dashed black line is for 2015-2025.

In the Florida Everglades and the Keys (Fig. 4.7), the record only begins in 2000, but the same pattern is also evident in these records. MHWs have become more frequent, and severe MHWs have occurred in 2015, 2019, and 2020 while the median number of MHW days per year has increased from less than 5 days to around 20 days. The majority of the stations in this region are

located in the Everglades and Florida Bay, with only 3 stations located in the Keys having more than 10 years of data.

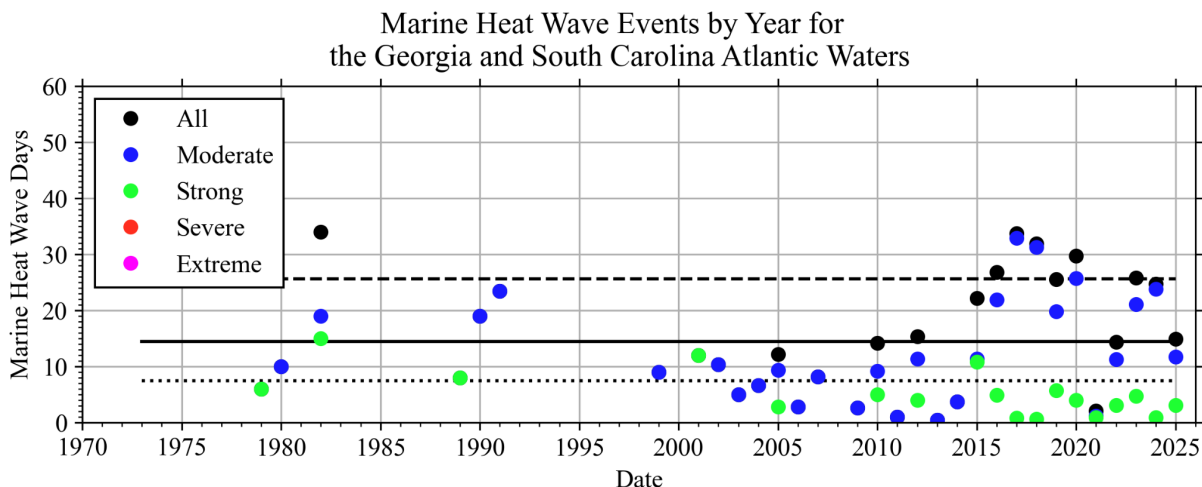


Figure 4.9: Mean Marine Heat Wave Days (black dots) per station by year in the Georgia and South Carolina Atlantic waters with the categories highlighted as moderate (blue), strong (green), severe (red), and extreme (pink). The solid black line represents the median marine heat wave days for 1975-1995; the dotted black line is for 1995-2015; and the dashed black line is for 2015-2025.

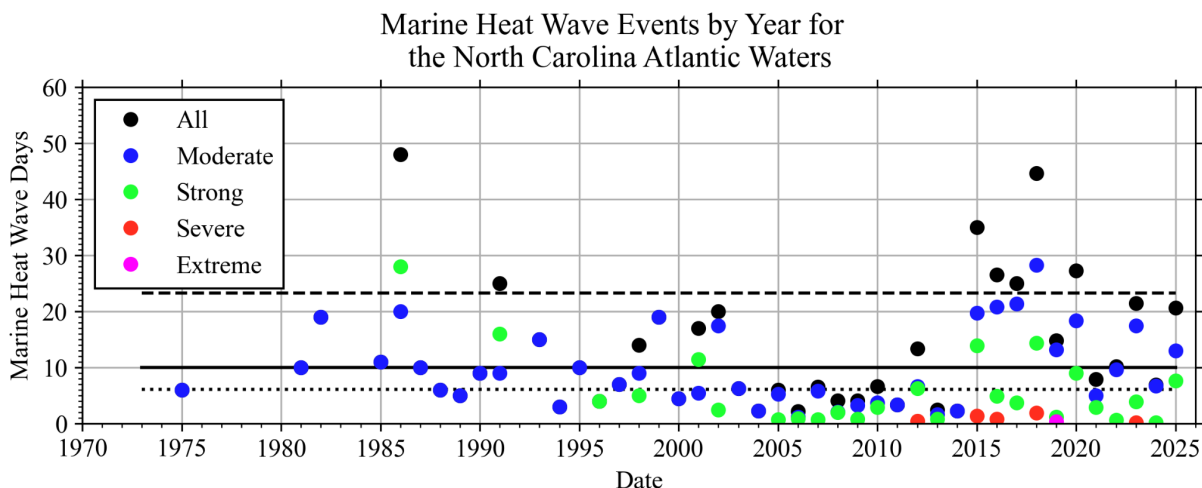


Figure 4.10: Mean Marine Heat Wave Days (black dots) per station by year in the North Carolina Atlantic waters with the categories highlighted as moderate (blue), strong (green), severe (red), and extreme (pink). The solid black line represents the median marine heat wave days for 1975-1995; the dotted black line is for 1995-2015; and the dashed black line is for 2015-2025.

The coastal waters east of Florida (Fig. 4.8) and Georgia and South Carolina (Fig. 4.9) repeat the same signals of an evolving climate, although the latter region had an anomalously warm start to

the 1990s. Neither region has experienced a severe or extreme MHW over their periods of SWT record, although consistent SWT tracking only began in the 2000s in each region. MHWs have increased from occurring 5-7 days per year (which is only one event per year) to occurring over 20 days per year. Before 2016, all data in this region come from two stations, and many of the stations added since then have been temporary arrays or riverine arrays.

The Atlantic waters off the coast of North Carolina are the longest, continuous SWT monitoring system in the SECOORA region, with consistent records back to 1985 and some stations having data from the 1970s. Over the 50 years of record, MHW days have increased from about 5 per year to over 20, as at other locations (Fig. 4.10). In this region, there have been 5 years with a severe MHW since 2012, and in 2019 there was a severe MHW.

## 5. Summary

This study analyzed sea water temperature (SWT) records across the coastal waters of the United States from Florida north to the Carolinas (the SECOORA region), with nearly 1,000 stations in the analysis. Among the stations included were NDBC partner buoys, USF-COMPS buoys on the West Florida Shelf, and typically bottom-mounted SWT probes from the FACT network.

All of these stations, when combined, provide a detailed image of the evolution of SWTs across the region. Over the 50 year period of record, temperatures have warmed at a rate greater than 0.3°C/decade. Accompanying the increasing temperatures is an increase in the occurrence and intensity of marine heatwaves (MHWs) across the region.

Full water column SWT measurements such as those from the West Florida Shelf COMPS buoys are essential for fully understanding the risks associated with MHWs, especially in regions where the water column features more than a single layer. With the increase in SWTs, many transient species are moving further north. Subsurface measurements are important in coastal shelf regions where there may be 2 or 3 distinct layers, even in shallower water, that simultaneously experience different categories of MHW.

Three regions—Pamlico Sound, the Florida Keys, and the Atlantic regions of Georgia and South Carolina—are significantly large regions that are lacking consistent, *in situ* SWT observations at the surface. The Pamlico Sound is significant as the second largest lagoon in the United States (Ross & Epperly, 1985) with significant nursery and fishing activities occurring in the Sound (NCDEQ, 2016; NCDMF, 2017). The Florida Keys contain approximately 350 km of coral reefs (Toth et al., 2022) that have experienced bleaching events in recent years (Ayad et al., 2023) due to increasing global SWTs (Yao & Wang, 2026). The Keys have recently begun routine *in situ* measurements of SWT, including many FACT probes, but these systems must remain in place for many years yet before there will be reliable climatologies for the region.

While all of the SECOORA region is at increasing risk from MHWs, the region most in need of a more comprehensive SWT monitoring system would be the Florida Keys, where the short-term FACT network measurements represent the only subsurface measurements. Coral reefs within this region are at increasing risk of bleaching events with warmer temperatures and the reefs are typically 2-15 meters below the surface. Such a sensor network is essential to understanding the occurrence of bleaching in this region.

Given the similarities between CORMP and COMPS and the importance of the Gulf Stream in the Carolinas with upwelling and downwelling in the coastal shelf region (Andres et al., 2023), a full array of water column measurements, including currents and salinity, is necessary to develop full understanding the regional physical processes over the course of seasons and years. Of most importance would be including measurements near the 10 meter isobath to capture upwelling and downwelling events.

With over 600 stations located across the SECOORA region, the FACT network represents one of the most important SWT measurements. These SWTs are especially important in regions with temperature-sensitive ecosystems that are being directly impacted by MHWs, such as the Florida Keys, Biscayne Bay, and other ecologically important regions where there are few measurement systems in place. With the FACT probes, data can be collected *post facto* to analyze temperature impacts in a region, whether it's coral bleaching, recruitment failure, or migration.

## References

- Ayad, M., C. M. Lee, J. W. Porter, V. Chirayath, C. L. Nivison, K. M. Vaughn, & R. Kudela. 2025. Impacts of the 2023 Marine Heatwave in the Florida Keys: Detection and Analysis of a Mass Coral Bleaching Event Using Spaceborne Remote Sensing Imagery. *Environmental Science & Technology*, **59**, 15227–15235, <https://doi.org/10.1021/acs.est.5c03122>
- Bushnell, M., R. Heitsenrether, J. Thomas, C. Galvarino, E. Burger, J. Dorton, & L. Leonard. 2018. Status and Near-Term Plans for the U.S. IOOS Quality Assurance / Quality Control of Real-time Oceanographic Data (QARTOD) Project. OCEANS 2018 MTS/IEEE Charleston, Charleston, SC, USA, 2018, 1–6, <https://doi.org/10.1109/OCEANS.2018.8604787>
- Cheung, W. W. L., T. L. Fröhlicher, V. W. Y. Lam, M. A. Oyinlola, G. Reygondeau, U. R. Sumaila, T. C. Tai, L. C. L. Teh, & . C. C. Wabnitz. 2021. Marine high temperature extremes amplify the impacts of climate change on fish and fisheries. *Science Advances*, **7**, eabh0895, <https://doi.org/10.1126/sciadv.abh0895>

- Eakin, C. M., H. P. A. Sweatman, & R. E. Brainard. 2019. The 2014–2017 global-scale coral bleaching event: insights and impacts. *Coral Reefs*, **38**, 539–545, <https://doi.org/10.1007/s00338-019-01844-2>
- Filbee-Dexter, K., T. Wernberg, S. P. Grace, J. Thormar, S. Fredriksen, C. N. Narvaez, C. J. Feehan, & K. M. Norderhaug. 2020. Marine heatwaves and the collapse of marginal North Atlantic kelp forests. *Scientific Reports*, **10**, 13388, <https://doi.org/10.1038/s41598-020-70273-x>
- Garrahou, J. et al. 2022. Marine heatwaves drive recurrent mass mortalities in the Mediterranean Sea. *Global Change Biology*, **28**, 5708–5725, <https://doi.org/10.1111/gcb.16301>
- Hobday, A. J., L. V. Alexander, S. E. Perkins, D. A. Smale, S. C. Straub, E. C. J. Oliver, J. A. Benthuyssen, M. T. Burrows, M. G. Donat, M. Feng, N. J. Holbrook, P. J. Moore, H. A. Scannell, A. Sen Gupta, & T. Wernberg. 2016. A hierarchical approach to defining marine heatwaves. *Progress in Oceanography*, **141**, 227–238, <http://doi.org/10.1016/j.pocean.2015.12.014>
- Hobday, A. J., E. C. J. Oliver, A. Sen Gupta, J. A. Benthuyssen, M. T. Burrows, M. G. Donat, N. J. Holbrook, P. J. Moore, M. S. Thomsen, T. Wernberg, & D. A. Smale. 2018. Categorizing and naming marine heatwaves. *Oceanography* **31**(2), 162–173, <https://doi.org/10.5670/oceanog.2018.205>
- Huang, B., V. F. Banzon, E. Freeman, J. H. Lawrimore, W. Liu, T. C. Peterson, T. M. Smith, P. W. Thorne, S. D. Woodruff, & H.-M. Zhang. 2015. Extended reconstructed sea surface temperature Version 4 [ERSSTv4]. Part I: upgrades and intercomparisons. *Journal of Climate*, **28**, 911–930, <https://doi.org/10.1175/JCLI-D-14-00006.1>
- Huang, B., P. W. Thorne, V. F. Banzon, T. Boyer, G. Chepurin, J. H. Lawrimore, M. J. Menne, T. M. Smith, R. S. Vose, & H.-M. Zhang. 2017. Extended Reconstructed Sea Surface Temperature, Version 5 (ERSSTv5): Upgrades, Validations, and Intercomparisons. *Journal of Climate*, **30**, 8179–8205, <https://doi.org/10.1175/JCLI-D-16-0836.1>
- Jones, T., J. K., Parrish, W. T. Peterson, E. P. Bjorkstedt, N. A. Bond, L. T. Ballance, V. Bowes, J. M. Hipfner, H. K. Burgess, & J. E. Dolliver. 2018. Massive Mortality of a Planktivorous seabird in response to a marine heatwave. *Geophysical Research Letters*, **45**, 3193–3202, <https://doi.org/10.1002/2017GL076164>
- Kennedy, J. J., N. A. Rayner, R. O. Smith, D. E. Parker, & M. Saunby. 2011. Reassessing biases and other uncertainties in sea surface temperature observations measured in situ since

- 1850: 2. Biases and homogenization. *Journal of Geophysical Research*, **116**, D14103, <https://doi.org/10.1029/2010JD015218>
- Kent, E. C., & P. G. Challenor. 2006. Toward estimating climatic trends in SST. Part II: random errors. *Journal of Atmospheric and Oceanic Technology*, **23**, 476–486, <https://doi.org/10.1175/JTECH1844.1>
- Lazarus, S. M., C. G. Calvert, M. E. Splitt, P. Santos, D. W. Sharp, P. F. Blottman, & S. M. Spratt. 2007. Real-time, high-resolution space-time analysis of sea surface temperatures from multiple platforms. *Monthly Weather Review*, **135**, 3158–3173, <https://doi.org/10.1175/MWR3465.1>
- Liu, Y., R. H. Weisberg, L. Sorinas, J. A. Law, & A. K. Nickerson. 2025. Rapid intensification of Hurricane Ian in relation to anomalously warm subsurface water on the wide continental shelf. *Geophysical Research Letters*, **52**, e2024GL113192, <https://doi.org/10.1029/2024GL113192>
- Mawren, D., J. Hermes, & C. J. C. Reason. 2022. Marine heat waves and tropical cyclones—two devastating types of coastal hazard in South-eastern Africa. *Estuarine, Coastal and Shelf Science*, **277**, 108056, <https://doi.org/10.1016/j.ecss.2022.108056>
- Moriarty, T., W. Leggat, S. F. Heron, R. Steinberg, & T. D. Ainsworth. 2023. Bleaching, mortality and lengthy recovery on the coral reefs of Lord Howe Island. The 2019 marine heatwave suggests an uncertain future for high-latitude ecosystems. *PLOS Climate*, **2**, e0000080, <https://doi.org/10.1371/journal.pclm.0000080>
- NCDEQ (North Carolina Department of Environmental Quality). 2016. North Carolina Coastal Habitat Protection Plan Source Document. North Carolina Division of Marine Fisheries, Morehead City, North Carolina, p. 475.
- NCDMF (North Carolina Division of Marine Fisheries). 2017. North Carolina License and Statistics Section Summary Statistics of License and Permit Program, Commercial Trip Ticket Program, NC Marine Recreational Information Program, Striped Bass Creel Survey in the Central and Southern Management Area, NC Recreational Saltwater Activity Mail Survey, Fisheries Economics Program. North Carolina Division of Marine Fisheries, Morehead City, North Carolina, p. 395.
- Nickerson, A.K., Weisberg, R.H., Zheng, L., Liu, Y. 2023. Sea surface temperature trends for Tampa Bay, West Florida Shelf and the deep Gulf of Mexico, *Deep-Sea Research Part II*, **211**, 105321, <https://doi.org/10.1016/j.dsr2.2023.105321>

- Nickerson, A. K., J. A. Zhang, R. H. Weisberg, & Y. Liu. 2025. Rapid intensification of Hurricane Ian (2022) in high shear, *Journal of Geophysical Research: Atmospheres*, **130**, e2024JD042024, <https://doi.org/10.1029/2024JD042024>
- Paerl, H. W., L. M. Valdes-Weaver, A. R. Joyner, & V. Winkelmann. 2007. Phytoplankton Indicators of Ecological Change in the Eutrophying Pamlico Sound System, North Carolina. *Ecological Applications*, **17**, S88–S101, <https://doi.org/10.1890/05-0840.1>
- Rathore, S, R. Goyal, B. Jangir, C. C. Ummenhofer, M. Feng, & M. Mishra. 2022. Interactions between a marine heatwave and tropical cyclone Amphan in the Bay of Bengal in 2020. *Frontiers in Climate*, **4**, 861477, <https://doi.org/10.3389/fclim.2022.861477>
- Reynolds, R. W., N. A. Rayner, T. M. Smith, D. C. Stokes, and W. Wang. 2002. An improved *in situ* and satellite SST analysis for climate. *Journal of Climate*, **15**, 1609–1625, [https://doi.org/10.1175/1520-0442\(2002\)015<1609:AISAS>2.0.CO;2](https://doi.org/10.1175/1520-0442(2002)015<1609:AISAS>2.0.CO;2)
- Reynolds, R. W., T. M. Smith, C. Liu, D. B. Chelton, K. S. Casey, & M. G. Schlax. 2007. Daily high-resolution blended analyses for sea surface temperatures. *Journal of Climate*, **20**, 5473–5496, <https://doi.org/10.1175/2007JCLI1824.1>
- Ross, S. W., & S. P. Epperly. 1985. Utilization of shallow estuarine nursery areas by fishes in Pamlico Sound and adjacent tributaries, North Carolina. In: Yanez-Arancibia, A. (Ed.), *Fish Community Ecology in Estuaries and Coastal Lagoons*, Ch. 10, 207–232. UNAM (Universidad Nacional Autonoma de Mexico) Press, México.
- Roxy, M. K., J. S. Saranya, A. Modi, A. Anusree, W. Cai, L. Resplandy, J. Vialard, & T. L. Fröhlicher. 2024. Future projections for the tropical Indian Ocean. In: C.C. Ummenhofer & R. R. Hood (Eds.), *The Indian Ocean and its Role in the Global Climate System*. Elsevier. pp. 469–482, <https://doi.org/10.1016/B978-0-12-822698-8.00004-4>
- Saranya, J. S., M. K. Roxy, P. Dasgupta, & A. Anand. 2022. Genesis and trends in marine heatwaves over the Tropical Indian Ocean and their interaction with the Indian summer monsoon. *Journal of Geophysical Research: Oceans*, **127**, e2021JC017427, <https://doi.org/10.1029/2021JC017427>
- Singh, V. K., M. K. Roxy, & M. Deshpande. 2021. Role of warm ocean conditions and the MJO in the genesis and intensification of extremely severe cyclone Fani. *Scientific Reports*, **11**, 3607, <https://doi.org/10.1038/s41598-021-82680-9>



- Slutz, R. J., S. J. Lubker, J. D. Hiscox, S. D. Woodruff, R. L. Jenne, D. H.. Joseph, P, M. Steurer, and J. D. Elms. 1985. Comprehensive ocean-atmosphere data set, release 1. NOAA Environmental Research Laboratories, Climate Research Program, Boulder, CO, 268 pp.
- Smale, D. A., T. Wernberg, E. C. J. Oliver, M. Thomsen, B. P. Harvey, S. C. Straub, M. T. Burrows, L. V. Alexander, J. A. Benthuisen, M. G. Donat, M. Feng, A. J. Hobday, N. J. Holbrook, S. E. Perkins-Kirkpatrick, H. A. Scannell, A. Sen Gupta, B. L. Payne, & P. J. Moore. 2019. Marine heatwaves threaten global biodiversity and the provision of ecosystem services. *Nature Climate Change*, **9**, 306–312, <https://doi.org/10.1038/s41558-019-0412-1>
- Smith, K. E., M. Aubin, M. T. Burrows, K. Filbee-Dexter, A. J. Hobday, N. J. Holbrook, N. G. King, P. J. Moore, A. Sen Gupta, M. Thomsen, T. Wernberg, E. Wilson, & D. A. Smale. 2024. Global impacts of marine heatwaves on shallow coastal foundation species. *Nature Communications*, **15**, 5052, <https://doi.org/10.1038/s41467-024-49307-9>
- Smith, K. E., M. T. Burrows, A. J. Hobday, N. G. King, P. J. Moore, A. Sen Gupta, M. S. Thomsen, T. Wernberg, & D. A. Smale. 2023. Biological Impacts of Marine Heatwaves. *Annual Review Marine Science*, **15**, 119-145, <https://doi.org/10.1146/annurev-marine-032122-121437>
- Smith, T. M., & R. W. Reynolds. 2002. Bias corrections for historical sea surface temperatures based on marine air temperatures. *Journal of Climate*, **15**, 73–87, [https://doi.org/10.1175/1520-0442\(2002\)015<0073:BCFHSS>2.0.CO;2](https://doi.org/10.1175/1520-0442(2002)015<0073:BCFHSS>2.0.CO;2)
- Smith, T. M., & R. W. Reynolds. 2003. Extended reconstruction of global sea surface temperatures based on COADS data (1854-1997). *Journal of Climate*, **16**, 1495–1510, [https://doi.org/10.1175/1520-0442\(2003\)016<1495:EROGSS>2.0.CO;2](https://doi.org/10.1175/1520-0442(2003)016<1495:EROGSS>2.0.CO;2)
- Smith, T. M., & R. W. Reynolds. 2005. A global merged land-air-sea surface temperature reconstruction based on historical observations (1880-1997). *Journal of Climate*, **18**, 2021–2036, <https://doi.org/10.1175/JCLI3362.1>
- Sorinas, L., Weisberg, R.H., Liu, Y., Law, J. (2023), Ocean-atmosphere heat exchange seasonal cycle on the West Florida Shelf derived from long term moored data. *Deep-Sea Research Part II*, **212**, <https://doi.org/10.1016/j.dsr2.2023.105341>
- Stuart-Smith, R. D., C. J. Brown, D. M. Ceccarelli, & G. J. Edgar. 2018. Ecosystem restructuring along the Great Barrier Reef following mass coral bleaching. *Nature*, **560**, 92–96, <https://doi.org/10.1038/s41586-018-0359-9>



- Toth, L. T., T. A. Courtney, M. A. Colella, S. A. Kupfner Johnson, & R. R. Ruzicka. 2022. The past, present, and future of coral reef growth in the Florida Keys. *Global Change Biology*, **28**, 5294–5309. <https://doi.org/10.1111/gcb.16295>
- Vose, R. S., D. Arndt, V. F. Banzon, D. R. Easterling, B. Gleason, B. Huang, E. Kearns, E. J. H. Lawrimore, M. J. Menne, T. C. Peterson, R. W. Reynolds, T. M. Smith, C. N. Williams, Jr., & D. B. Wuretz. 2012. NOAA's merged land-ocean surface temperature analysis. *Bulletin of the American Meteorological Society*, **93**, 1677-1685, <https://doi.org/10.1175/BAMS-D-11-00241.1>
- Wernberg, T., D. A. Smale, F. Tuya, M. S. Thomsen, T. J. Langlois, T. de Bettignies, S. Bennett, & C. S. Rousseaux. 2013. An extreme climatic event alters marine ecosystem structure in a global biodiversity hotspot. *Nature Climate Change*, **3**, 78–82, <https://doi.org/10.1038/nclimate1627>
- Woodruff, S. D., H. F. Diaz, J. D. Elms, & S. J. Worley. 1998. COADS release 2 and metadata enhancements for improvement of marine surface flux fields. *Physics and Chemistry of the Earth*, **23**, 517–526, [https://doi.org/10.1016/S0079-1946\(98\)00064-0](https://doi.org/10.1016/S0079-1946(98)00064-0)
- Young, J. M., M. E. Bowers, E. A. Reyier, D. Morley, E. R. Ault, J. D. Pye, R. M. Gallagher, & R. D. Ellis. 2020. The FACT Network: Philosophy, Evolution, and Management of a Collaborative Coastal Tracking Network. *Marine and Coastal Fisheries: Dynamics, Management, and Ecosystem Science*, **12**, 258–271, <https://doi.org/10.1002/mcf2.10100>
- Yao, Y., & C. Wan. 2026. Record-breaking marine heatwaves across global coral reefs in 2024. *Geophysical Research Letters*, **53**, e2026GL122086, <https://doi.org/10.1029/2026GL122086>

### Supplementary Information

This section includes figures that support Figs. 4.6-4.10 in the body of the report. These plots show the number of stations within each of the subregions within the SECOORA region for the purpose of showing the sparseness of the data record until the past 10 to 20 years.

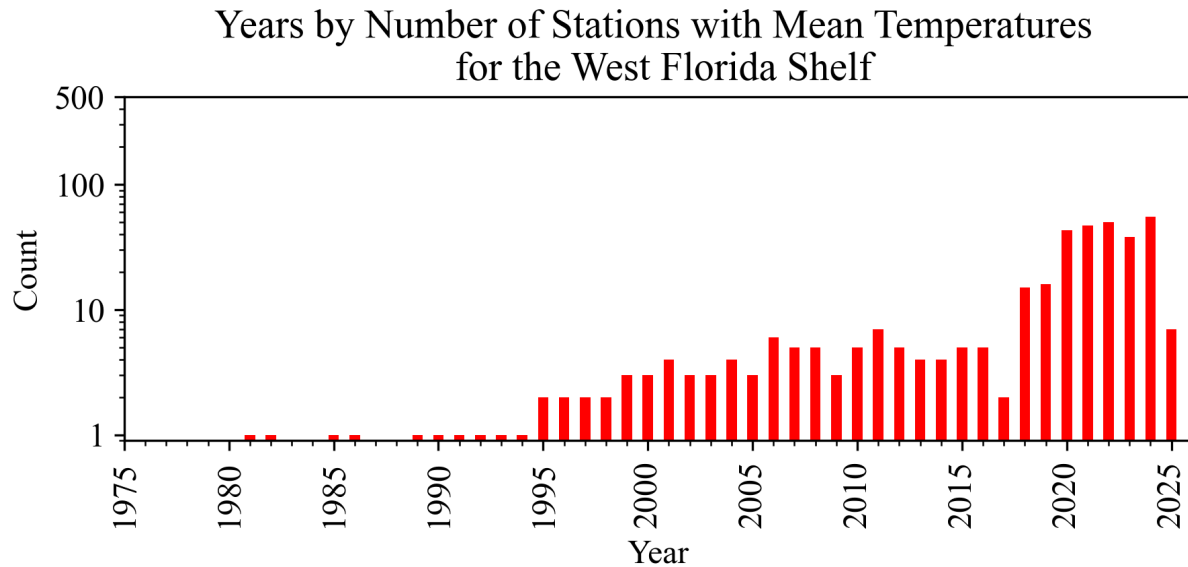


Figure S1: Bar chart showing the number of stations measuring SWTs on the West Florida Shelf in each year.

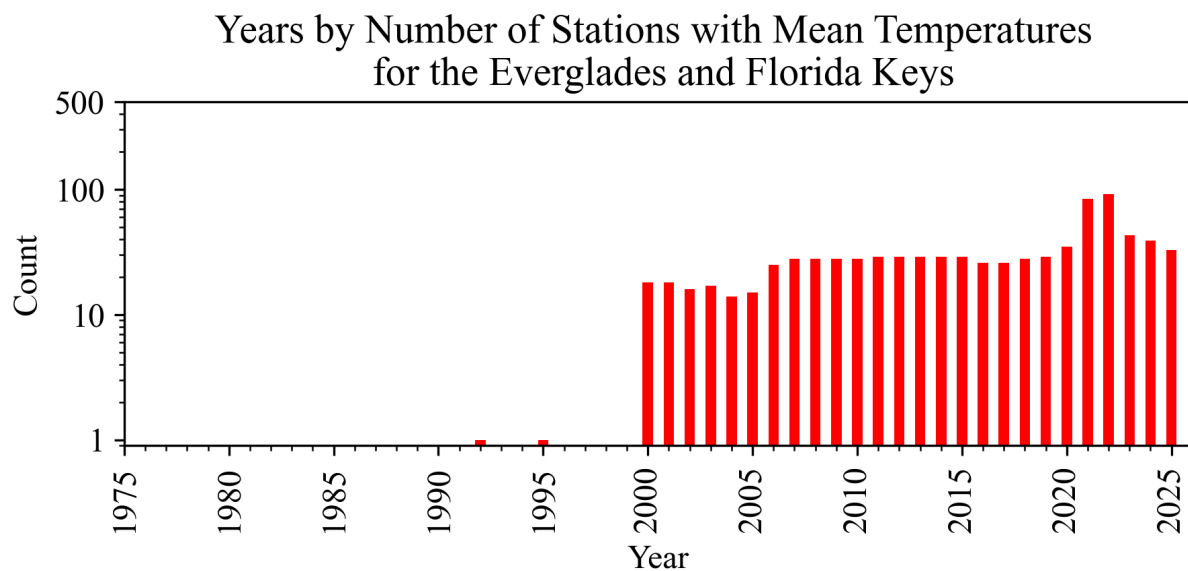


Figure S2: Bar chart showing the number of stations measuring SWTs within the Florida Everglades and Florida Keys in each year.

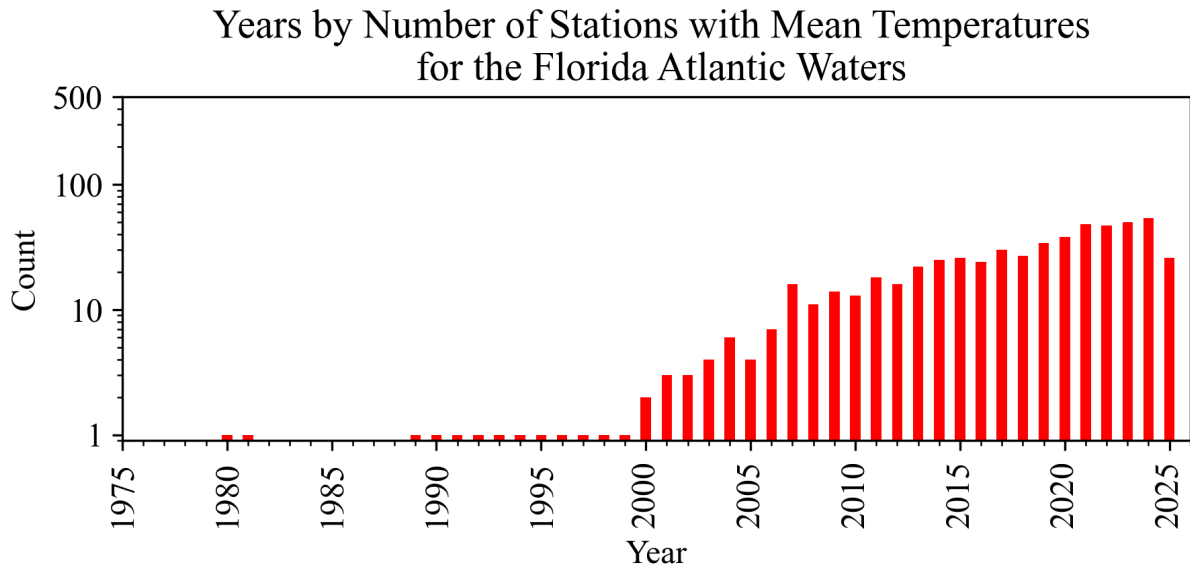


Figure S3: Bar chart showing the number of stations measuring SWTs within the Florida Atlantic waters for each year.

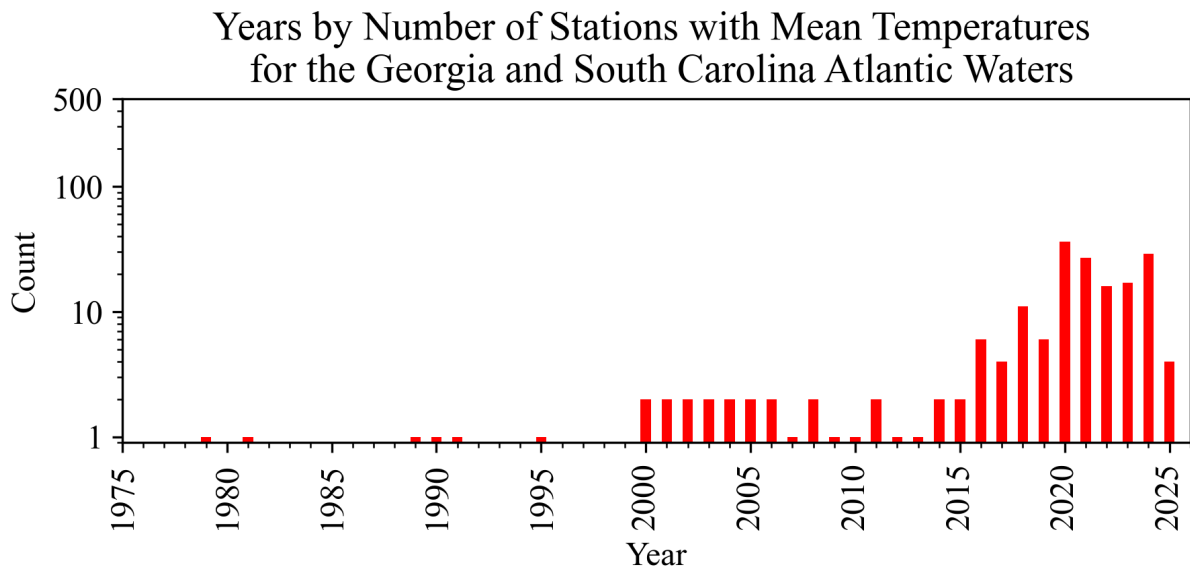


Figure S4: Bar chart showing the number of stations measuring SWTs within the Georgia and South Carolina Atlantic waters in each year.

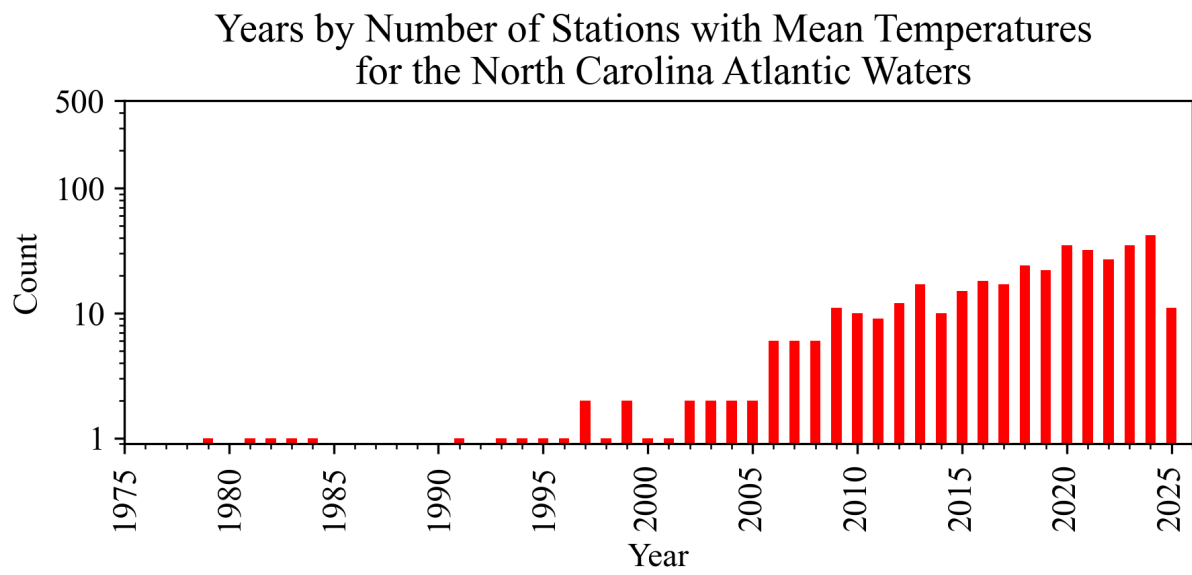


Figure S5: Bar chart showing the number of stations measuring SWTs within the North Carolina Atlantic waters in each year.